Using FaIR to explore the global warming projections under different emission scenarios

Contribution

Sry Wei is the only author for this project. I set up the required conda environment to run the Finite Amplitude Impulse Response simple climate model (FaIR), learned the model framework from the basic_run_example.ipynb, coded in the cmip6_ssp_emissions_run.ipynb for my own experiments. I performed three numerical experiments, produced figures, interpreted results, and wrote the final paper.

Introduction

FaIR is a reduced complexity model and used as an emulator to the GCMs, for it can quickly test climate-response questions while retaining the main physical constraints, which can be modified by five dimensions, including “time”, “scenario”, “config”, “specie”, and “box/layer”. I used FaIR to explore how global mean surface temperature changes under different emission scenarios and focused on answering two questions: (1) can FaIR reproduce the overall projection patterns shown in class figures, and (2) how much does projected warming change if we only modify one emissions pathway?

Methods

The three experiments shared the same model configuration: the Thornhill (2021) methane lifetime scheme, annual time steps from 1750 to 2100, the default full-climate species list from read_properties(), scenario inputs loaded from RCMIP, and CMIP6-calibrated climate-response parameter sets from “4xCO2_cummins_ebm3.csv”. Different FaIR configurations represent different climate-response assumptions and therefore provide an ensemble spread, which was summarized by using the ensemble mean and the 5th–95th percentile range.

The three experiments used the same workflow as the cmip6_ssp_emissions_run.ipynb official example, with scenario inputs from RCMIP, the default full-climate species set, and CMIP6-calibrated climate-response configurations. Different FaIR configurations represent different climate-response assumptions, so I summarized the output using the ensemble mean and the 5th–95th percentile range. In this setup, the model uses prescribed scenario pathways to generate projected temperature responses, and the simplified energy-balance structure allows the response to evolve gradually rather than instantaneously.

For Experiment 1, I reproduced the class projection pattern shown in the AR5-based reference figure (Fig. 1b) using SSP pathways instead of the RCP pathways used in the reference. My FaIR result is shown in Fig. 1a. I ran FaIR under SSP1-2.6, SSP2-4.5, SSP4-6.0, and SSP5-8.5 and plotted global mean surface temperature anomalies relative to 1986–2005. This was designed as a FaIR analogue of the course slide based on IPCC AR5 Figure 12.5, which shows global annual mean surface air temperature anomalies relative to 1986–2005 from CMIP5.

For Experiment 2, I used SSP2-4.5 as the baseline case and created a modified scenario test by reducing CO_2 fossil-fuel-and-industry (FFI) emissions by 50% starting in 2030, leaving the rest of the configurations unchanged. This was implemented directly inside cmip6_ssp_emissions_run.ipynb by editing the CO2 FFI emissions field after the baseline SSP data were loaded (Fig. 2).

For Experiment 3, I used the five AR6 priority scenarios (SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5) and computed long-term warming as the 2081–2100 average relative to 1995–2014. This matches the warming metric used in IPCC AR6 Figure 4.11, panel (d), and allows a visual comparison shown in Fig. 3. I then compared the resulting long-term warming

values with the AR6 assessed values reported in Table 4.5 in Fig. 4, providing a separate quantitative comparison.

Results

Experiment 1: Reproducing scenario divergence

The first experiment successfully reproduced the broad scenario-divergence pattern shown in the course reference figure (Fig. 1b). In all scenarios, warming trajectories remain relatively close in the early part of the century and diverges later By 2100, the FaIR ensemble mean warming relative to 1986–2005 was 1.57 K for SSP1-2.6, 2.88 K for SSP2-4.5, 3.45 K for SSP4-6.0, and 5.52 K for SSP5-8.5. The uncertainty ranges become wider with time, especially under stronger forcing. Higher forcings also produce larger late-century warming.

Figure 1a and Figure 1b shared a common historical pathway before diverging, but my FaIR experiment is not an exact replication of the AR5 figure we saw in class. The AR5 reference figure (Fig. 1b) is based on CMIP5 concentration-driven RCP experiments and extends to 2300. My experiment (Fig. 1a) uses FaIR with SSP scenarios and ends at 2100 (example notebook time setup limit). In addition, the shaded range in Figure 1a represents the 5th–95th percentile range across my FaIR ensembles results. The shaded range in the AR5 reference figure (Fig. 1b) represents the 5–95% range across the distribution of individual CMIP5 models. The two uncertainty ranges evaluated from different sources should not be interpreted equivalently.

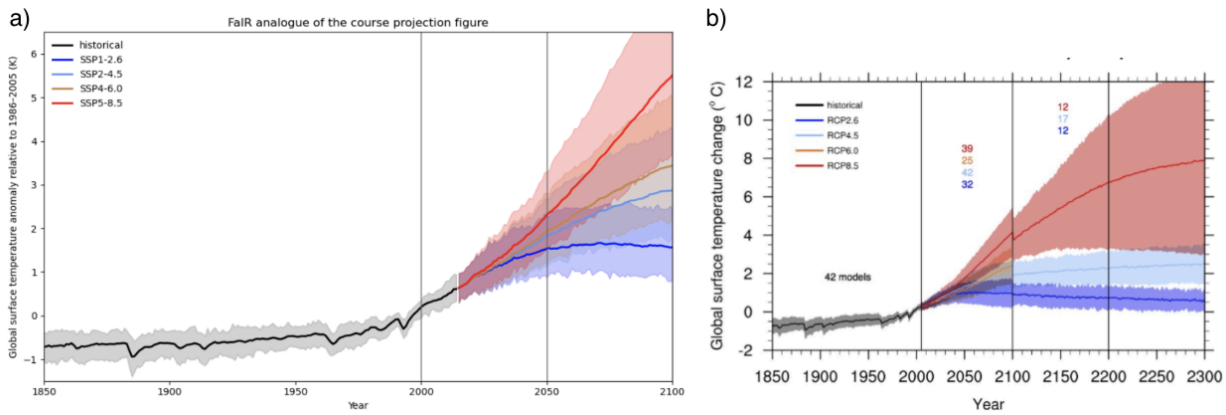

Figure 1. Comparison between my FaIR analogue of the average global warming projection and the IPCC AR5 reference figure. (a) FaIR-simulated global surface temperature anomalies relative to 1986–2005 for SSP1-2.6, SSP2-4.5, SSP4-6.0, and SSP5-8.5. Lines show ensemble means, and shading shows the 5th–95th percentile range. (b) Course reference figure based on IPCC AR5 Figure 12.5. The FaIR result reproduces the overall scenario-divergence pattern, but it is not an exact replication because it uses SSP scenarios and FaIR ensemble spread rather than CMIP5 RCP projections.

Experiment 2: Reduced fossil-fuel CO2 emissions in SSP2-4.5

The perturbation experiment produced an expected result. Before 2030, the modified trajectory and the baseline SSP2-4.5 trajectory are identical because we haven't applied the perturbation. After 2030, the modified case shows less warming than the baseline case. By the late century, the separation is clear in the ensemble mean, although the uncertainty ranges still overlap.

Lower CO2 FFI emissions reduce the amount of carbon entering the atmosphere, which lowers radiative forcing and therefore reduces long-term warming. The response is gradual rather than instantaneous because CO2 accumulates over time and because the climate system has thermal inertia in my current setup. This experiment changes only one emissions term rather than

constructing a fully self-consistent mitigation pathway, indicating that FaIR is a useful tool for emission change tests and sensitivity experiments compared to the computation-heavy GCMs.

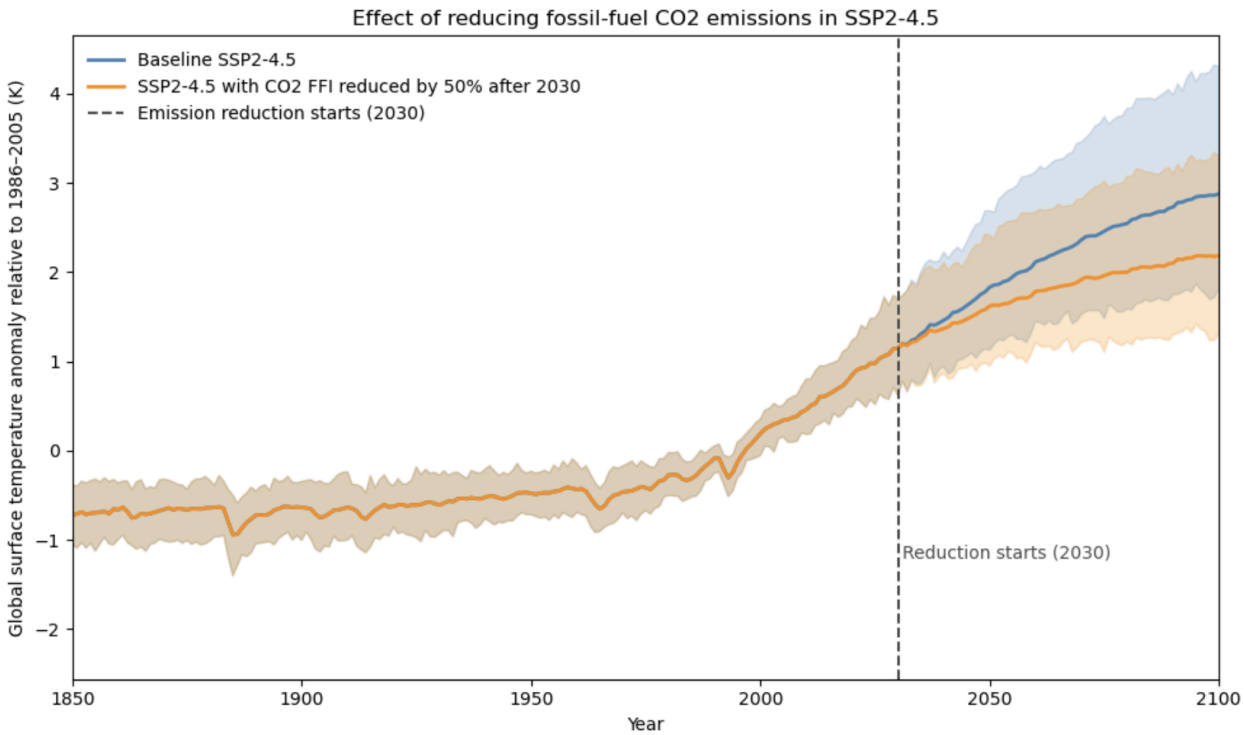

Figure 2. Effect of reducing fossil-fuel-and-industry (FFI) CO2 emissions in SSP2-4.5. The perturbed experiment reduces CO2 FFI by 50% beginning in 2030. Lines show ensemble means, shading shows the 5th–95th percentile range, and the dashed line marks the start of the perturbation. The reduced-emissions case shows expected lower late-century warming than the baseline SSP2-4.5 case.

Experiment 3: Comparison with IPCC AR6

The third experiment gave good qualitative agreement with IPCC AR6 Figure 4.11(d). I used the same AR6 priority scenarios and the long-term warming metric. My FaIR results (Fig. 3a) reproduced the same scenario ordering as the AR6 reference figure (Fig. 3b). The

low-emissions scenarios flatten more strongly later in the century, while the high-emissions scenarios continue to warm. The spread also becomes wider in stronger forcing scenarios.

However, I found a systematic warm bias relative to AR6 assessed values in the quantitative comparison (Fig. 4). For 2081–2100 relative to 1995–2014, my FaIR means were 0.80 K, 1.30 K, 2.45 K, 3.59 K, and 4.60 K for SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5, respectively, compared with AR6 best estimates of 0.6 K, 0.9 K, 1.8 K, 2.8 K, and 3.5 K. The difference increases from about 0.20 K in SSP1-1.9 to about 1.10 K in SSP5-8.5. The corresponding AR6 best estimates were 0.6 K, 0.9 K, 1.8 K, 2.8 K, and 3.5 K. These values were calculated from my FaIR analysis and are reproducible from the project code(github link).

This mismatch is not surprising because my results are based on direct FaIR runs using the official example notebook workflow and configurations, while the IPCC AR6 Figure 4.11(d) is an assessed product. According to the AR6 figure caption, AR6 Figure 4.11(d) is constructed by averaging constrained CMIP6 ranges and emulator ranges rather than plotting a single raw emulator output. In that sense, the systematic warm bias is likely due to the differences between my direct FaIR setup and the AR6 assessed workflow, including differences in configuration choices and uncertainty construction.

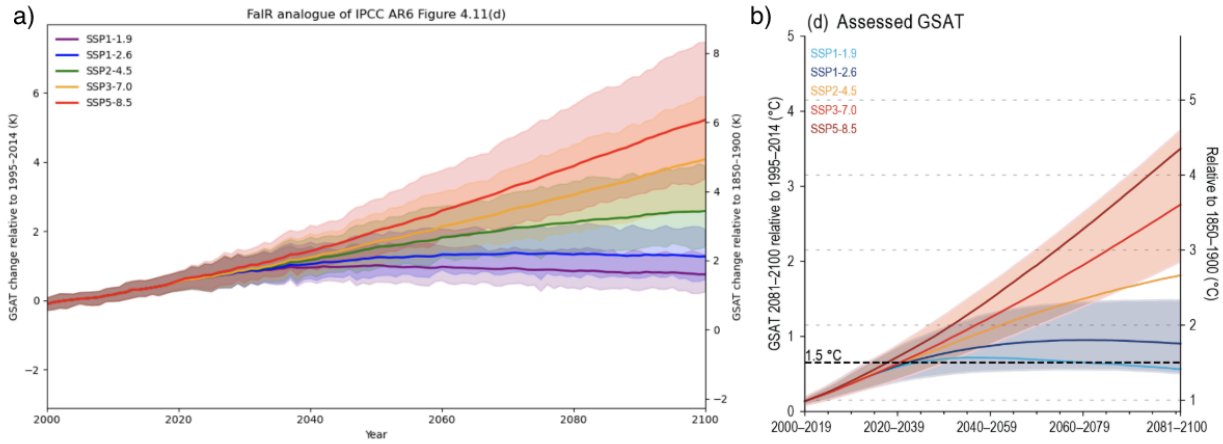

Figure 3. Comparison between FaIR projections and IPCC AR6 Figure 4.11, panel (d). (a) FaIR-simulated global surface air temperature (GSAT) change relative to 1995–2014 for SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5. Lines show ensemble means, and shading shows the 5th–95th percentile range. (b) IPCC AR6 Figure 4.11, panel (d), shown for reference. FaIR reproduces the overall scenario ordering and late-century separation seen in AR6.

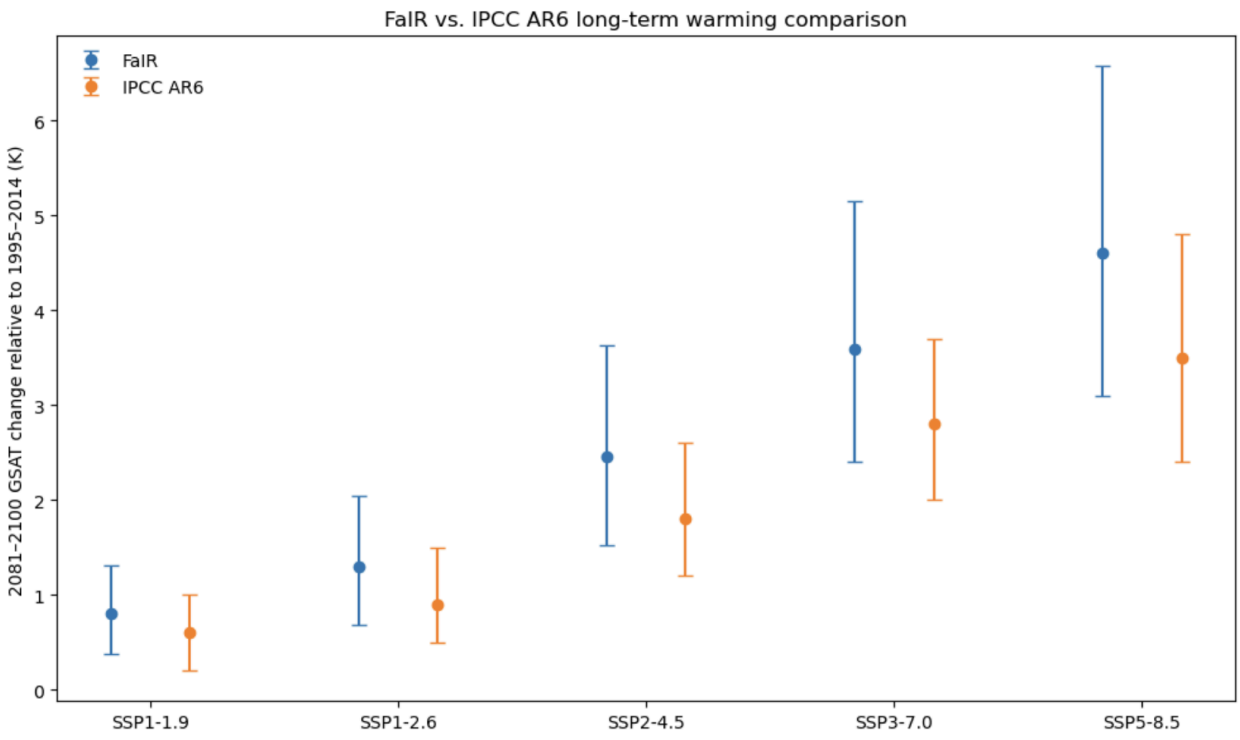

Figure 4. Quantitative comparison of long-term warming between FaIR and IPCC AR6 Table 4.5. Points show mean warming over 2081–2100 relative to 1995–2014, and error bars show uncertainty ranges. FaIR values are generally warmer than the AR6 assessed values, with larger differences under higher-forcing scenarios.

Conclusion

Overall, this project shows that FaIR is able to capture the main qualitative structure of future global warming under SSP scenarios. It reproduces the expected projection patterns with

higher-forcing pathways leading to greater late-century warming. It provides a convenient tool to test how temperature responds to modified emissions. The three experiments were also consistent in results. Experiment 1 reproduced the broad scenario pattern seen in the course reference figure, Experiment 2 showed reduced warming after the imposed CO2 modification, and Experiment 3 matched AR6 qualitatively. Together, these experiments demonstrate that FaIR can capture the main qualitative patterns of future warming while also providing a flexible framework for simple emissions sensitivity tests.

In addition, the direct FaIR runs were systematically warmer than the AR6 assessed values in my current setup. My results are based on the official example notebook workflow and configurations, whereas the AR6 comparison uses an assessed framework with a different uncertainty construction. This suggests that my current setup captures the overall scenario structure but does not exactly reproduce the AR6 assessment workflow.

There are also several limitations. First, the analysis was adapted from the official example notebooks rather than built as a fully independent model implementation. Second, the perturbation experiment changed only CO2 FFI and therefore should be interpreted as a simple sensitivity test rather than a realistic mitigation scenario. Third, the uncertainty ranges in my FaIR results are not exactly the same as the assessed values shown in the AR6 figure. The next step would be to investigate the source of the systematic warm bias, test other emissions perturbations (CH_4, aerosols, etc), and explore the configuration setup details for customized FaIR runs.

References

IPCC, 2013: Climate Change 2013: The Physical Science Basis. Working Group I contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change, Chapter 12, Figure 12.5.

IPCC, 2021: Lee, J.-Y., J. Marotzke, G. Bala, et al., 2021: Future Global Climate: Scenario-Based Projections and Near-Term Information. In Climate Change 2021: The Physical Science Basis. Working Group I contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Chapter 4, Figure 4.11 and Table 4.5.

Github: